Mock exam 2

Level: Medium to hard **Marks: 25** **Time: 45 minutes**

- All questions are compulsory.
- The marks for each question are shown against it.
- In Section A choose the one correct option. In the other sections write complete sentences.

Section A: Multiple choice questions

6 x 1 = 6 marks

1. A bar magnet is broken into three pieces. How many poles do the three pieces have in all? [1]

(a) 2 **(b)** 3

(c) 4 **(d)** 6

2. An iron bar is hung freely from a thread and turned gently several times. Each time, it will [1]

(a) rest along the north-south direction **(b)** rest along the east-west direction

(c) rest along any direction **(d)** keep on rotating

3. The North pole of a bar magnet is slowly brought near the North pole of a compass needle. The North pole of the needle [1]

(a) moves closer to the magnet **(b)** moves away from the magnet

(c) does not move **(d)** starts spinning

4. On some magnets the poles are not marked N and S. On such a magnet, the North pole may be shown by [1]

(a) a white dot **(b)** blue paint

(c) a black line **(d)** a small hole

5. **Assertion (A):** An iron bar brought near a compass needle cannot push either end of the needle away. [1]
Reason (R): Repulsion takes place only between the like poles of two magnets.

(a) Both A and R are true, and R explains A.

(b) Both A and R are true, but R does not explain A.

(c) A is true, but R is false.

(d) A is false, but R is true.

6. A strong magnet is brought near a coin. The coin is not attracted at all. Which conclusion is correct? [1]

(a) The coin is made of a non-magnetic material.

(b) The coin is made of iron.

(c) The magnet has lost one of its poles.

(d) The coin is a magnet.

Section B: Very short answer questions

2 x 2 = 4 marks

7. A steel paper clip lies at the bottom of a plastic bottle filled with water. How can you take it out without wetting your hand? Why does this work? [2]

8. Sana dips a U-shaped magnet into a heap of iron filings and lifts it. Where will most of the filings stick? What does this tell her about the magnet? [2]

Section C: Short answer questions

2 x 3 = 6 marks

9. Dev has a bar magnet with marked poles and two unmarked metal bars, P and Q. [3]
One end of bar P is attracted by the North pole of the magnet and also by its South pole.
One end of bar Q is attracted by the North pole of the magnet but repelled by its South pole.
(a) Which bar is a magnet? (b) Which pole is at that end of the bar? (c) What is the other bar?

10. (a) Why can a freely suspended magnet be used to find directions? [3]
(b) Why does it behave in this way?
(c) Why is one end of a compass needle usually painted red?

Section D: Long answer question

1 x 5 = 5 marks

11. Kabir has an iron sewing needle, a bar magnet, a small piece of cork, a bowl and some water. [5]

(a) Write the steps he should follow to make a magnetic compass. (3 marks)

(b) How can he check that the needle has become a magnet? (1 mark)

(c) He turns the cork gently a few times. What should he see each time if his compass works? (1 mark)

Section E: Case-based question

1 x 4 = 4 marks

12. Read the passage and answer the questions. [4]

Four ring magnets P, Q, R and T are put on a wooden rod, one above the other. P is at the bottom. Q floats above P without touching it. R sits on Q and touches it. T floats above R without touching it. The top face of P is a North pole.

(i) Which pole is the bottom face of Q? Give a reason.

(ii) Which pole is the top face of Q?

(iii) Which pole is the bottom face of R? Give a reason.

(iv) Which pole is the bottom face of T?